Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering (PCCoE)

(An Autonomous Institute)

Affiliated to Savitribai Phule Pune University(SPPU)

Subject:- Object Oriented Modelling and System Design [BCER27PC02]

FA1 Instructions

FA Activity :- Design to Database – OOMSD Presentation

Instructions for Students

1. **Group Formation**
   - Form a group of **2–3 students**.
   - Each group must work on **one system/case study**.
2. **Case Study Selection**
   - Select your **own system/case study** based on a real-world application.
   - The selected system should have sufficient scope to identify **multiple objects, interactions, and database entities**.
   - **Topics must not be repeated** by different groups.
   - All group members must ensure that their **names, roll numbers, and selected topic** are correctly entered in the sheet.

     https://docs.google.com/spreadsheets/d/16BS7iybBuyrTKJU8peKhBDouxNbbckT2XlFpTrEDr10/edit?usp=sharing

3. **Design Sequence**
   - Prepare your system design in the following sequence:
     Case Study → CRC Cards → Class Diagram → Use Case Diagram → Sequence Diagram → ER Diagram → Normalization
4. **CRC Cards**
   - Identify important **classes/objects**.
   - Define the **responsibilities** of each class.
   - Identify **collaborators**.
   - Apply appropriate **GRASP principles and SOLID**, wherever applicable.
5. **UML Class Diagram**
   - Include **classes, attributes, and methods**.

- Show appropriate **relationships and multiplicity**.
- Ensure that the classes identified in the CRC cards are properly reflected in the Class Diagram.

6. **Use Case Diagram**
   - Identify the **actors** and major system functionalities.
   - Show appropriate relationships such as **include** and **extend**, wherever applicable.
7. **Sequence Diagram**
   - Select **one important use case** from your Use Case Diagram.
   - Show the interaction between the **actor and system objects**.
   - Clearly represent the **sequence of messages/operations**.
   - The objects used in the Sequence Diagram should be consistent with your Class Diagram.
8. **ER Diagram & Normalization**
   - Identify **entities, attributes, and relationships**.
   - Clearly specify **Primary Keys and Foreign Keys**.
   - Convert the design into **relational tables**.
   - Apply **normalization up to 3NF**, wherever applicable.
9. **Presentation**
   - Each group will present the **complete system design**.
   - Presentation duration: **8–10 minutes per group**.
   - A short **viva/question-answer session** will follow the presentation.
   - Every member should be able to **explain and justify the complete design**.
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10. **Live Demonstration of Diagrams**

- During the presentation, students must **open the actual UML modelling tool** used to create their diagrams, such as **draw.io, StarUML, or Lucidchart**.
- Keep the **presentation slides and modelling tool open side by side**.
- Explain the diagrams **directly from the modelling tool** rather than only showing screenshots.

11. **Tools**

- Students may use **draw.io, StarUML, Lucidchart**, or any other suitable UML modelling tool.
- Diagrams must be **clear, properly labelled, and follow standard UML notation**.

Sr. No.	Assessment Criteria	Expected Performance	Marks
1	**CRC Cards**	Correct identification of classes, responsibilities and collaborators; appropriate application of GRASP	**2**
2	**UML Class Diagram**	Correct classes, attributes, methods, relationships and multiplicity	**2**
3	**Use Case Diagram**	Correct actors, use cases and appropriate relationships	**2**
4	**Sequence Diagram**	Correct actor-object interaction, messages and sequence for selected use case	**2**
5	**ER Diagram & Normalization**	Correct entities, relationships, keys and normalization up to 3NF	**2**
	Total		**10**